

STRATEGIC BUSINESS PORTFOLIO NAVIGATOR

(USING BCG MATRIX AND PYTHON)

MILESTONES :

➤ Accomplished:

- Data Manipulation in Excel to find Relative Market Shares and Market Growth.
- Data Interpretation using Pandas and formation of BCG MATRIX using Matplotlib.
- Strategic Analysis of BCG Matrix formed.

➤ Partially Accomplished:

- The implementable Strategy for each Product is crisp, it could have been in an elaborative way and unique for each.

➤ Failed to Accomplish:

- The competitor sales are user defined it should have been automatically taken once the market is selected, this I was unable to achieve.

REASON OF CHOOSING THE PROJECT AND HOW THE PROBLEM IS SOLVED:

There are many small-scale industries which donot follow a proper strategy to check their relative performance and they endup taking wrong decisions.

This is a major cause of declining small-startups and industries in our country.

This tool solves their problem by simply collecting their standard business unit's data of sales and volumes and uses the concept of business strategy developer proposed by the Boston Consulting Group (BCG Matrix).

Being small scale startups their market size is limited so they here can choose their competitors themselves as they need to give the sales of their competitors as well.

Through this simple programming tool and basic excel maintenance the industries can analyse their business portfolio accurately and can take decisions that would benefit and help grow their

business and also prevent them from entering into any kind of bankrupt situations.

HOW TO RUN:

Pre requisites:

- 1.Any code editor with installed python and its modules like tkinter, pandas , matplotlib,
- 2.Excel File saved in CSV format in the computer which has data of sales (both current and past years) along with sales of competitors which the user considers and volumes product-wise sold.

(Note: The Manipulation in Excel is not done by code it is already done in stored csv file)

Once, you have all the pre requisites, you are ready to go.

Manipulation in Excel:

In the model excel file (already stored in csv format) the following manipulations have been done:

- Relative Market Shares:

A column named market_share is created which basically calculates the market share rate of the product using the formula:

Market Share=Current Sales/Maximum of Competitor and your present sales

- **Market Growth Rate:**

A column named growth_rate is created which basically calculates the market growth rate of the product using the formula:

Market growth rate=ABS value of (Current Sales-Past sales)/Past Sales

By this Relative Market Share will be always less than 1 Radius:

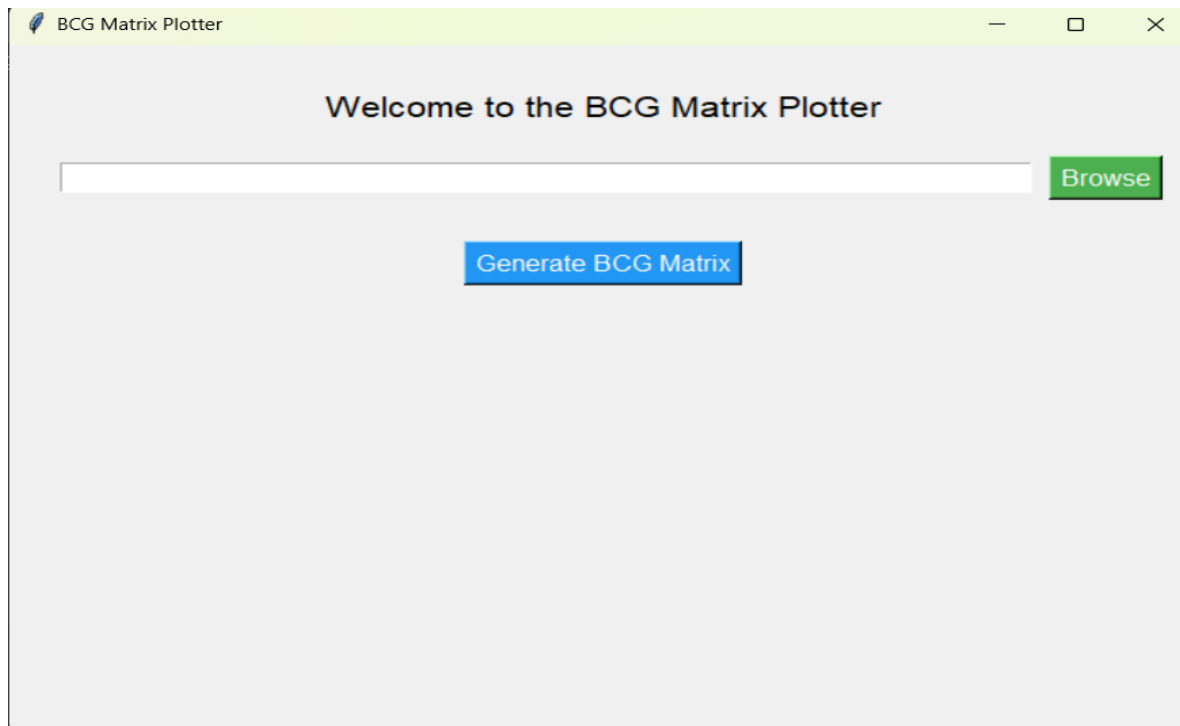
The Volumes of products should be entered in a new column named radius and that should be normalized by dividing volumes by 10 power number of digits in volumes entered.

Running the Code and How it Works:

Once you are done with all the Manipulations in excel and saved it in csv format, you have to open

your code editor like (VS Code) and access the python file .

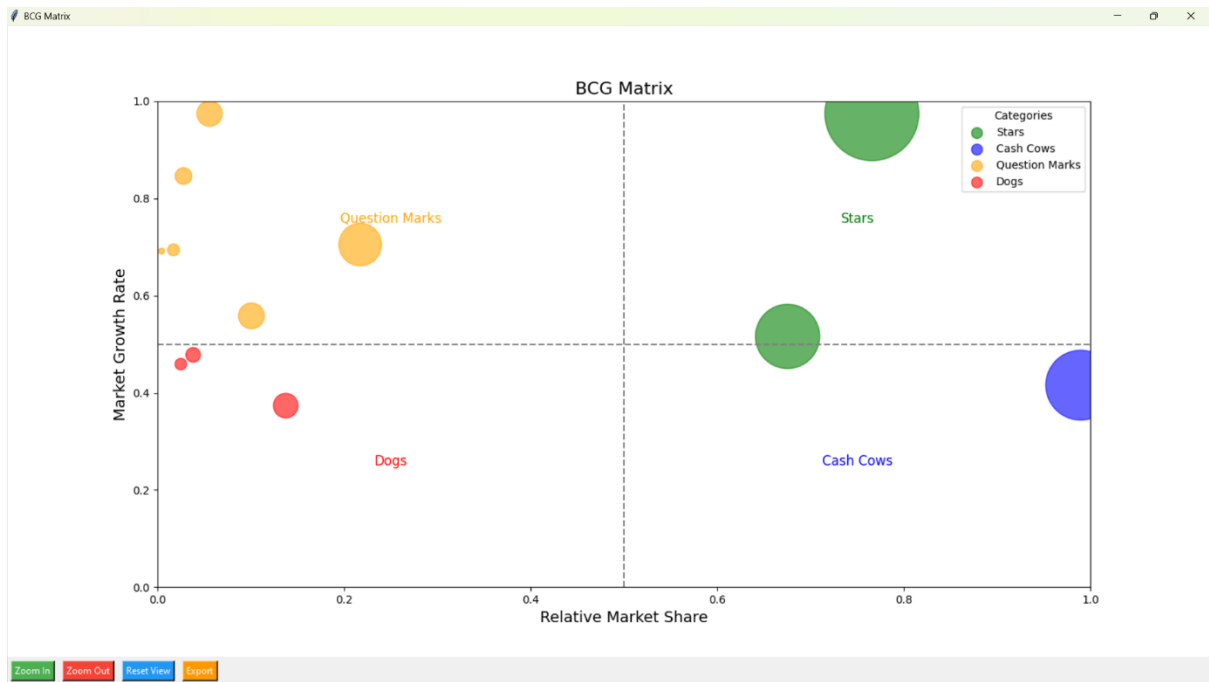
Now, just run the Python File, you will be able to see a window like:



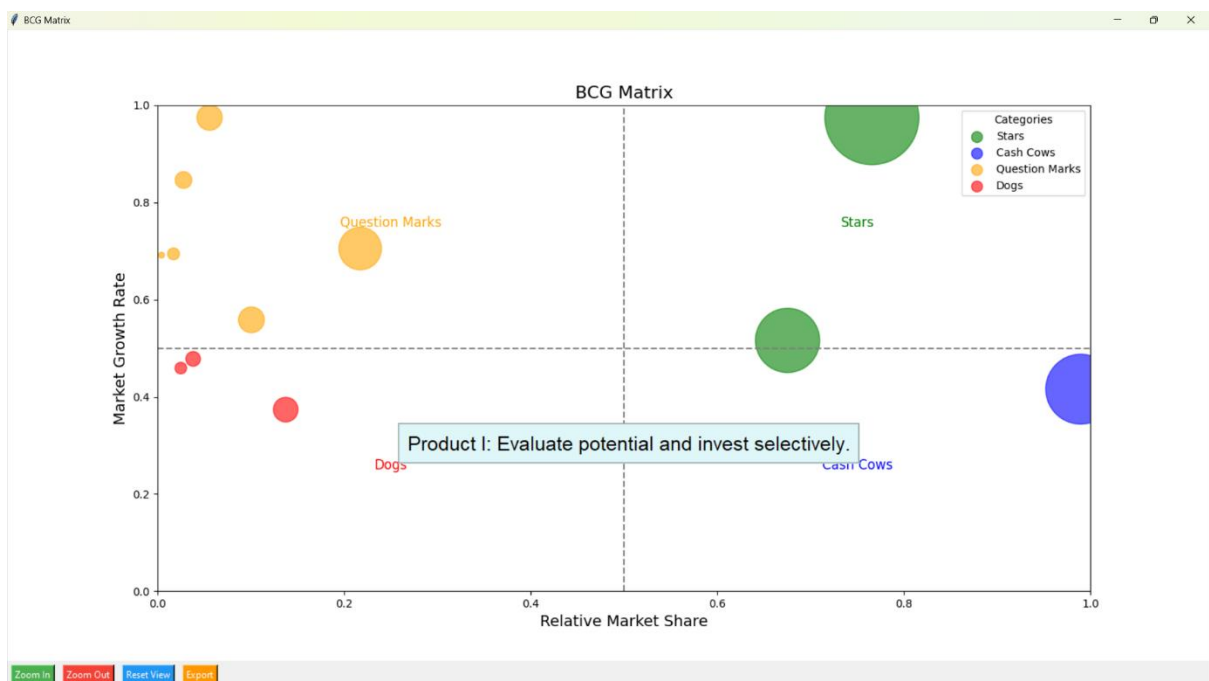
Click on Browse Button to select your CSV File.

Then click on Generate BCG MATRIX.

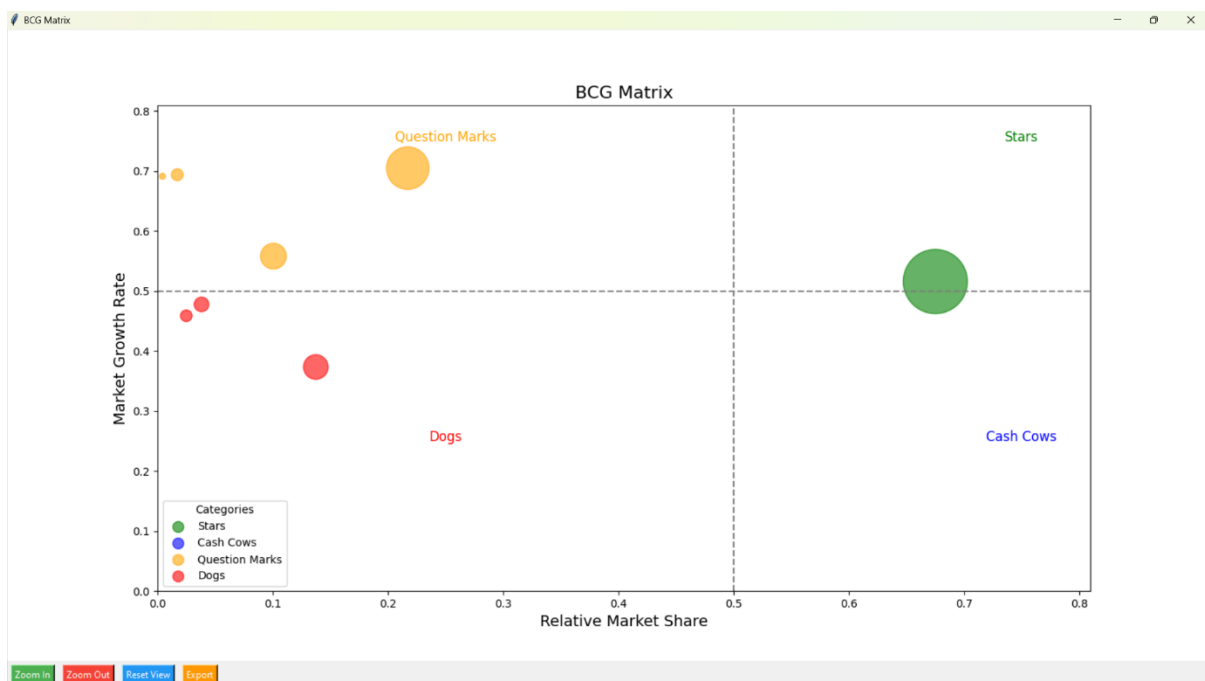
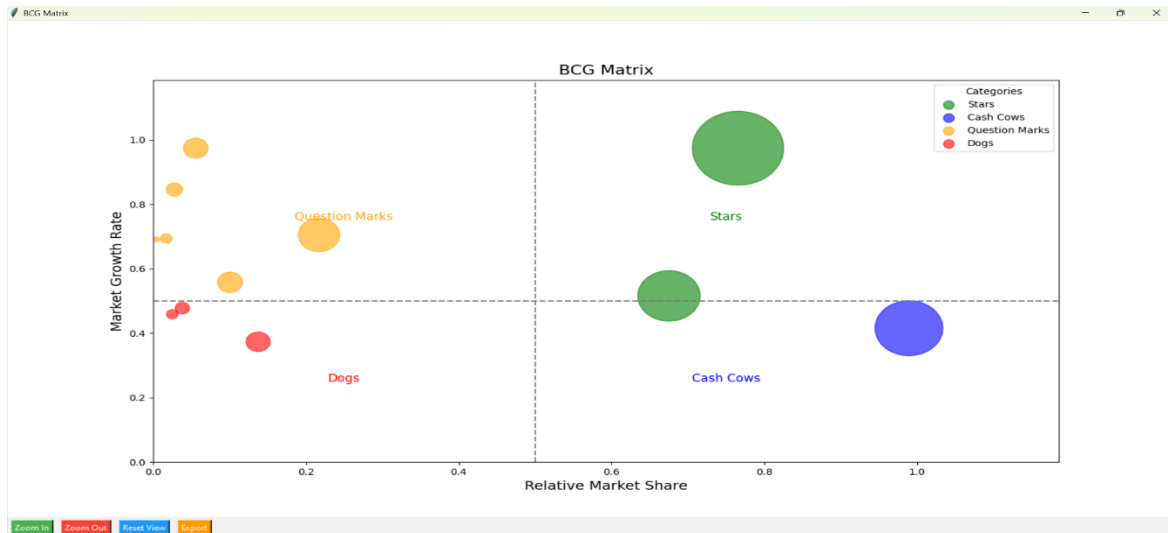
The BCG MATRIX will be generated as shown:



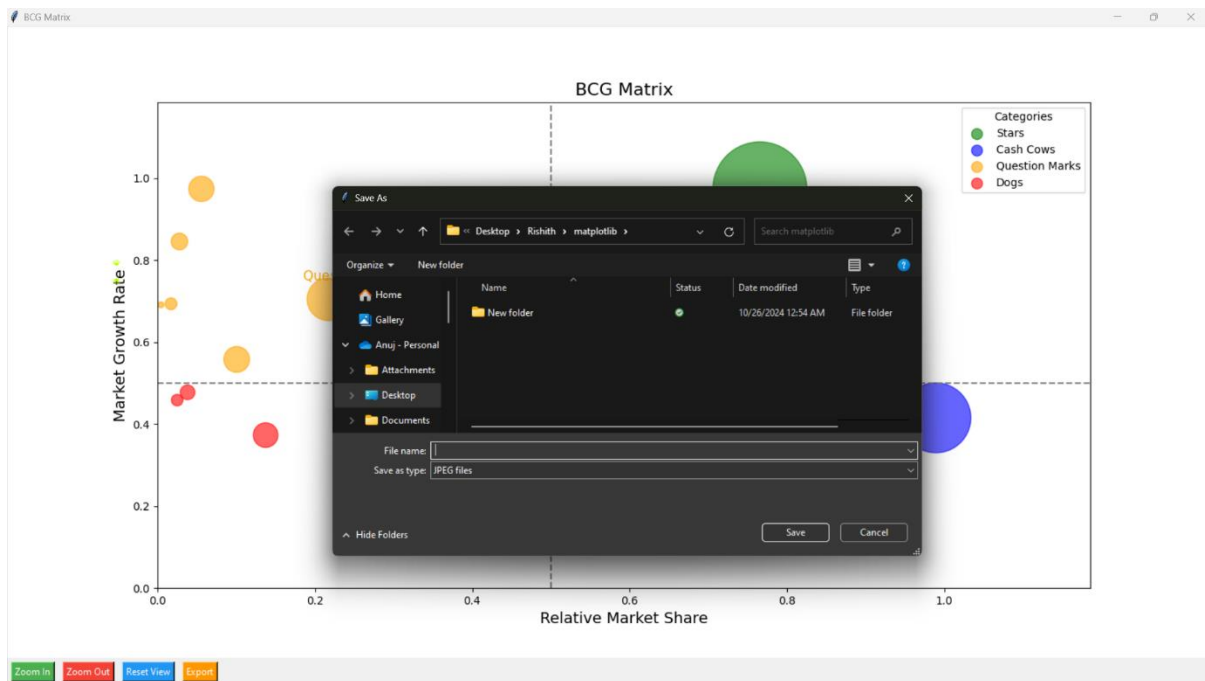
Now to know the strategies of individual Products you can hover your mouse on the products in the BCG Matrix and you will be able to get the output of strategy as:



You can also use Zoom-In and Zoom-Out buttons to know access detailed parts of Matrix:



You can also export the BCG matrix formed to jpeg and PDF formats and store it your desired location using the export button Provided.



Code Explanation and Implementation File:

The Program uses the following Modules in Python:

1. Matplotlib
2. Pandas
3. Tkinter

In the code I have created various functions for the various features of the Project. The flow and list of functions is as follows:

1. Function to classify the products into 4 categories (Stars, Question Marks, Cash Cows,

Dogs) based on the values of growth and market rates.

2. Function to use plot the BCG matrix by using the matplotlib-properties, which includes creating the initial window followed by creating the plot setting the limits and adding labels.
3. The in the function I used the tkinter properties to create the windows and added up the buttons with required positioning and styling.
4. Along with these this function also includes functions created for each feature like zoom-in, zoom-out and export.
5. Then after this function there is a function to open the csv function and another function to read the csv file using pandas.
6. After this there is a main window setup made using tkinter which includes all the labelling and styling involved in setting up the windows.

The overview can be understood by the flowchart given:

